Question:

Patrick has three glue sticks that are partially used. One has 1/6 left, the second has 2/3 left and the third one has 1/2 left. If a glue stick is 12 millimeters long originally, what is the total length of the glue sticks that are not used?

Reasoning + Answer:

<think> To find the total length of the glue sticks that are not used, we need to first determine the length of the glue that is used in each stick. The first stick has 1 / 6 left, so it has 5 / 6 of the original length used. 5 / 6 of 12 millimeters is (5 / 6) * 12 = 10 millimeters. The second stick has 2 / 3 left, so it has 1 / 3 of the original length used. 1 / 3 of 12 millimeters is (1 / 3) * 12 = 4 millimeters. The third stick has 1 / 2 left, so it has 1 / 2 of the original length used. 1 / 2 of 12 millimeters is (1 / 2) * 12 = 6 millimeters. Now, we need to add the lengths of the used glue in each stick to find the total length of the glue that is used. Total length of used glue = 10 + 4 + 6 = 20 millimeters. Since the original length of the glue sticks is 12 millimeters each, the total length of the glue sticks that are not used is 3 * 12 - 20 = 24 millimeters. </think><answer> 24</answer>